

Timur Rizvanov

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Education

Boston University

Bachelor of Arts in Chemistry and Computer Science (Double Major)

Boston, MA

Expected May 2027

Related Graduate Coursework: Advanced Optimization Algorithms; GPU Programming in CUDA/C++; Molecular Quantum Mechanics III.

Related Undergraduate Coursework: Advanced Organic Chemistry 1, 2; Quantum Mechanics; Instrumental Analysis Laboratory.

Research Experience

Bravaya Lab - Boston University

Machine Learning & Computational Chemistry Researcher - Protein Electron Transfer

Boston, MA

September 2025 - Present

- Built deep-learning models that predict electronic couplings between protein residues in seconds instead of expensive QM calculations; trained on 270,000+ Q-Chem calculations (38 residue-pair types), with median cross-validated sMAPE of 10% across residue-pair types and ~99% sign accuracy.
- Combined these models with physics-based energy calculations (Marcus theory) in a development version of the lab's eMap software, so it can quickly calculate protein electron-transfer.
- Built an automated screening pipeline (sequence clustering, genome-neighborhood filtering, AlphaFold 3 modeling, electron-transfer analysis) that narrowed 18,000+ bCCP-superfamily sequences to 11 candidate MauG-like enzymes for experimental characterization by the Elliott group.
- Lead theoretical modeling in an ongoing Galagan Lab collaboration to improve an electrochemical nicotine biosensor (NicA2 electron transfer).

Vajda Lab - Boston University

Computational Chemistry Researcher - ML-Guided Docking

Boston, MA

September 2024 - March 2025

- Acquired, cleaned and standardized the protein-ligand structure datasets used to train and test a graph-neural-network docking method.
- Wrote Python benchmarking scripts that scored more than 3,500 predicted docking models against reference structures to measure accuracy.
- Built automated structural-QC checks flagging steric clashes and geometric inconsistencies in predicted complexes to diagnose model failures.

Dries Lab - Boston University Chobanian & Avedisian School of Medicine

Computational Biology Researcher - Scientific Software

Boston, MA

September 2024 - May 2026

- Lead author of dbVisuals (R package, not yet released) and author/maintainer of its engines [rDeckgl](#) and [rMosaic](#) (CRAN): database-backed, GPU-accelerated interactive visualization, 30x+ faster than ggplot2 and plotly WebGL, handling 20 million points on a laptop.
- Co-developed [dbverse](#), a database-backed framework that runs larger-than-memory analysis of millions of cells on ordinary computers, orders of magnitude faster than in-memory and file-backed methods.

Prior Research: Foster Research Group, September 2023 - May 2024 (Cancer-Targeting Nanoparticle Synthesis); CADD Research Group, June 2022 - May 2024 (Computational Chemistry, Data Analysis and Dataset Preparation).

Engineering & Leadership

MacroGlide

Co-Founder & Chief Technology Officer (CTO)

Boston, MA

June 2024 - Present

- Created the company's project-management system and have led a team of 5+ people for over 1.5 years.
- Developed automated AI content pipelines for social media: the [main Instagram account](#) grew to 450,000+ followers; now runs 50+ accounts.
- Developed an automated [newsletter platform](#) (multi-source data acquisition, agentic news selection and writing with automated quality scoring, email delivery infrastructure); 5,000+ new subscribers monthly.
- Automated internal engineering and project-management processes, roughly tripling the team's engineering output.

Publications & Manuscripts

- E. C. Ruiz, V. Jarzabek, J. Chen, **T. Rizvanov**, I. Amin, R. Dries. "dbverse scales spatial omics analysis with embedded analytical databases." *bioRxiv*, 2026. DOI: [10.64898/2026.08.09.743742](https://doi.org/10.64898/2026.08.09.743742)
- A. J. Kranc, **T. Rizvanov**, S. J. Elliott, K. B. Bravaya. "Unearthing Family Secrets: Searching for Shared Electron Transfer Pathways in the bCCP Superfamily." Submitted, 2026; [presented at ACS Fall 2026 \(COMP\)](#).
- **T. Rizvanov**, E. C. Ruiz, R. Dries. "dbVisuals: hardware-accelerated and interactive visualizations for spatial omics in R." In preparation, 2026.
- **T. Rizvanov**, K. B. Bravaya. Deep-learning prediction of electronic couplings for protein electron transfer. Manuscript in preparation, 2026.

Technical Skills

Machine Learning: Python, PyTorch, Deep Learning (ANN, GNN), Dataset Curation and Preparation, Model Benchmarking, LLM Agents.

Computational Chemistry: Q-Chem, DFT, Molecular Dynamics, Marcus Theory, Statistical Thermodynamics, AlphaFold 3, Docking.

Scientific Computing & HPC: C/C++, CUDA, GPU Optimization, Linux, HPC Cluster Computing (SGE), Git.

Data & Software Engineering: SQL, PostgreSQL, DuckDB, Apache Arrow, Parquet, R (CRAN Packages), TypeScript, CI/CD.

Laboratory Chemistry: Organic Synthesis, Retrosynthesis, Nanoparticle Synthesis, NMR, IR, HPLC, LC-MS, Column Chromatography.

Awards

- Outstanding Research Award, ACS Fall 2026 National Meeting, Division of Computers in Chemistry (COMP).
- Winner, [ClawBio Hackathons: Agentic Genomics](#), 2026.
- Three-time winner, National Stage of Chemistry Olympiad, 2021-2023; International Chemistry Olympiad (IChO) Qualifier participant, 2023.